

LYMPHOID NEOPLASIA

IL-16 production is a mechanism of resistance to BTK inhibitors and R-CHOP in lymphomas

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KEY POINTS

- IL-16 secretion drives BTK inhibitor and R-CHOP resistance; targeting IL-16/CD9 restores treatment response in B-cell lymphomas.
- Serum IL-16 may help identify patients at risk of BTK inhibitor failure and guide use of IL-16/CD9-targeted therapies in lymphomas.

Resistance to Bruton tyrosine kinase (BTK) inhibitors remains a major clinical challenge in B-cell lymphomas and often occurs in the absence of BTK or PLCG2 mutations. Here, we investigated nongenetic mechanisms of ibrutinib resistance in marginal zone lymphoma (MZL) and their broader therapeutic implications. Continuous ibrutinib exposure generated a resistant MZL model that showed cross-resistance to BTK inhibitors and degraders, without evidence of multidrug resistance or genetic alterations. Integrated transcriptomic, epigenetic, and proteomic analyses revealed extensive reprogramming, including the activation of phosphatidylinositol 3-kinase (PI3K)/AKT, MAPK, and MYC pathways, repression of apoptosis and oxidative phosphorylation, and a prominent cytokine-secretory phenotype. Interleukin-16 (IL-16) emerged as a central mediator of resistance. IL-16 was transcriptionally upregulated, actively secreted, and sufficient to induce ibrutinib resistance across multiple CD9⁺ models of MZL, mantle cell lymphoma, chronic lymphocytic leukemia (CLL), and activated B-cell-like diffuse large B-cell lymphoma. Serum IL-16 levels were elevated in patients with ibrutinib-refractory CLL without BTK/PLCG2 mutations. Mechanistically, IL-16 engaged CD9-enriched membrane microdomains to activate PI3Kδ, thereby sustaining AKT and extracellular signal-regulated kinase signaling, stabilizing MYC, inducing NF-κB-dependent programs, and upregulating antiapoptotic effectors, including BCL2A1. Pharmacological or genetic disruption of the IL-16/CD9/PI3K axis restored sensitivity to BTK inhibitors and R-CHOP (rituximab, cyclophosphamide, hydroxydaunorubicin, vincristine [Oncovin], and prednisone) and abrogated IL-16-induced signaling in primary CLL samples. In conclusion, an IL-16/CD9-driven, epigenetically regulated survival pathway represents one possible mechanism of resistance to BTK inhibitors and chemoimmunotherapy, supporting therapeutic targeting of this axis in refractory B-cell lymphomas.

Introduction

The introduction of pharmacological inhibitors of Bruton tyrosine kinase (BTK) into clinical practice has been among the most successful advances in treating patients with lymphoid tumors and autoimmune disorders.¹⁻⁴ BTK is downstream of

multiple receptors expressed and active in normal and neoplastic B cells, including the B-cell receptor (BCR), chemokine receptors (eg, CXCR4), and Toll-like receptors. Covalent and noncovalent inhibitors, and more recently, proteolysis-targeting chimeras, have shown strong preclinical and clinical activity in patients with chronic lymphocytic

leukemia (CLL), follicular lymphoma, mantle cell lymphoma (MCL), activated B-cell–like diffuse large B-cell lymphoma (ABC DLBCL), and marginal zone lymphoma (MZL).^{1,2,5}

Although BTK inhibitors have transformed the treatment landscape for B-cell malignancies, they are not curative. Some patients exhibit primary resistance, whereas others relapse or progress despite initial responses.^{1–3} We and others have reported genetic and nongenetic resistance mechanisms,^{1,3,6,7} yet these findings do not fully account for the high interpatient and inpatient clinical variability observed. To further elucidate resistance mechanisms, we developed an ibrutinib-resistant cell line derived from a patient with splenic MZL, harboring somatic mutations in *NOTCH2* and *KLF2*, hallmarks of the “NNK” (NF- κ B/NOTCH/KLF2) splenic MZL cluster.⁸ This model identified interleukin-16 (IL-16) as a key driver of acquired resistance, conferring survival advantages through CD9-dependent activation of NF- κ B and phosphatidylinositol 3-kinase (PI3K)–AKT signaling.

Materials and methods

Cell culture, generation of drug-resistant models, treatments, and drug-response assays

B-cell lymphoma cell lines were cultured under the recommended conditions as previously described.^{9,10} Media were supplemented with fetal bovine serum, antibiotics, and L-glutamine. Ibrutinib-resistant derivatives were generated by continuous exposure to the 90% inhibitory concentration (IC₉₀) of ibrutinib (15 μ M) until stable resistance developed; parental controls were cultured in parallel without drug. Biological replicates were generated by splitting resistant cultures 1 month after resistance acquisition and maintaining them independently for 6 months. Cell line identity was confirmed by short tandem repeat profiling, and all cultures were routinely tested for mycoplasma negativity. Selected cell lines were stably transduced with mCherry and monitored using the IncuCyte S3 live-cell imaging system. Small-molecule inhibitors, recombinant human IL-16, a neutralizing anti-IL-16 antibody, and a CD9-binding peptide were obtained from commercial sources. Drug responses were assessed using real-time IncuCyte imaging, with cell growth quantified as percent confluence. Selected experiments were validated using MTT (3-(4,5-dimethylthiazol-2-yl)-2,5-diphenyltetrazolium bromide) assays after 72 hours of treatment. IC₅₀ values were calculated using 4-parameter logistic models, and drug interactions were evaluated using Chou-Talalay, MuSyC, HSA, and ZIP synergy models. Primary CLL cells were isolated from peripheral blood by density-gradient centrifugation and purified by flow cytometry. Effects of treatments were evaluated after 4 hours of exposure to compounds using a colorimetric MTT assay to assess cell viability. Phospho-flow cytometry was used to assess basal and stimulated signaling responses following BCR engagement. Detailed methods are provided in the supplemental Materials and methods, available on the *Blood* website.

Flow cytometry, immunoblotting, and cytokine profiling

Surface and intracellular protein expression was analyzed by flow cytometry using standard fixation and permeabilization

protocols. Immunoblotting was performed to assess the expression and phosphorylation of key signaling proteins, including BTK, AKT, SYK, RELB, and antiapoptotic effectors, using antibodies listed in the supplemental Tables. Conditioned media were collected after 72 hours and analyzed using human cytokine arrays. Serum IL-16 levels were measured by Luminex assay in samples obtained from patients with ibrutinib-treated CLL enrolled under institutional review board–approved protocols with informed consent. Detailed methods are provided in the supplemental Materials and methods.

Genomic, epigenomic, transcriptomic, and proteomic analyses and data mining

Whole-exome sequencing, RNA sequencing (RNA-seq), small RNA-seq, DNA methylation profiling, and targeted DNA sequencing were performed as previously described.^{9,11} Differential expression and pathway analyses were conducted using established pipelines, with significance defined by false discovery rate–adjusted thresholds.

Global proteomic profiling was performed by liquid chromatography–tandem mass spectrometry using data-independent acquisition. Protein identification and quantification were conducted using DIA-NN, followed by normalization, imputation, and differential expression analysis in R. Chromatin accessibility was assessed by ATAC (assay for transposase-accessible chromatin) sequencing using the Omni-ATAC protocol. Sequencing data were processed using nf-core pipelines, and differential accessibility and transcription factor enrichment analyses were performed using established bioinformatics tools.

Statistical analyses were performed in R. Unless otherwise stated, 2-sided tests were used, and adjusted $P < .05$ were considered significant. Detailed methods are provided in the supplemental Materials and methods.

OpenAI’s ChatGPT was used to support exploratory data mining of author-generated data sets (eg, organizing variables, proposing grouping/annotation schemes, and drafting code-free summaries of patterns).

Results

Development of ibrutinib-resistant cells in MZL

The VL51 cell line was exposed to increasing concentrations of the BTK inhibitor ibrutinib for 6 months until stable resistance was acquired. The developed ibrutinib-resistant model (VL51-Ibru) had an IC₅₀ 15-fold higher than that of parental cells (PAR).

IC₅₀ calculation was validated after 3 weeks of culture in a drug-free medium. Resistant cells showed reduced sensitivity to covalent and noncovalent BTK inhibitors and BTK degraders (supplemental Figure 1). The multidrug resistance phenotype was evaluated by maintaining sensitivity to the multidrug resistance substrate vincristine: no increased expression of multidrug resistance 1/2 was observed (supplemental Figure 2).

We then studied potential changes in the morphology of VL51-Ibru cells compared with their parental counterpart using

electron microscopy. Ibrutinib-resistant cells were smaller than PAR but with more filopodia/membrane protrusions (supplemental Figure 3A). In addition to the more complex membrane, mitochondria were rounder, larger, and more fragmented in resistant cells than in PAR, suggesting greater mitochondrial activation in VL51-Ibru (supplemental Figure 3B).

Finally, VL51-Ibru did not show significant changes in sensitivity to PI3K inhibitors (idelalisib, duvelisib, copanlisib, and umbralisib) when compared with PAR (supplemental Figure 4).

Upregulation and secretion of IL-16 in VL51 IBR-RES cells

No DNA copy number variations or mutations were detected by whole exome sequencing or targeted sequencing of genes recurrently mutated in lymphomas (including *BTK* and *PLCG2*) in resistant cell lines compared with PAR (supplemental Table 3), indicating a nongenetic mechanism of resistance. Despite similar DNA profiles, resistant cells exhibited distinct transcriptomic and methylation profiles compared with their parental counterparts (Figure 1A). At the transcriptome level, VL51-Ibru showed overexpression of genes encoding proteins involved in cytokine secretion (*IL16*, *LCP2*, and *CXCL10*), immune response (*FCER1A*, *CD300*, and *CSF2*), cell proliferation (*CCNB1* and *MKI67*), integrins (*ITGAM* and *ITGA11*), and members of the PI3K-AKT (*PIK3R6*, *GPER1*, and *PRR5L*), MAPK (*LRRK2*, *MAP2*, and *RASGRP4*), and NF- κ B (*TNF*, *LTA*, and *CD44*) pathways, as well as genes related to resistance to targeted therapies in lymphoma (*EGR1/2*, *IGF1*, and *BCL2A1*). VL51-Ibru also showed a lower expression of tumor suppressor genes (*CDKN2B*, *CDKN2A*, and *CDKN1A*), negative regulators of PI3K-AKT-mTOR (*PIK3IP1* and *DEPTOR*), genes located at 7q31, frequently deleted in SMZL (*TASR3*, *CAV1*, and *DNAJB9*),^{12,13} and apoptosis-related genes (*PDCD4* and *ANXA1*) (Figure 1A; supplemental Table 4).

Integration of transcriptome and DNA promoter methylation profiles of VL51-Ibru cells and PAR identified upregulated, unmethylated genes, including secreted factors such as IL-16 and tumor necrosis factor (TNF) and PI3K-AKT signaling genes (*PIK3R6* and *RASGRP4*). Comprehensive analyses of small RNA and RNA profiles identified upregulated microRNAs (miRNAs; miR-5585, let-7c, and miR-1244) and 2 repressed miRNAs that target IL-16 (ie, miR-125a and miR-4451)¹⁴ (Figure 1A; supplemental Table 4). Gene set enrichment analyses revealed enrichment of genes involved in cell cycle, PI3K/AKT, MAPK, and NF- κ B signaling pathways, actin polymerization, and IL-16-associated pathways in VL51-Ibru. In contrast, PAR were characterized by elevated antigen presentation, p53 signaling, and IRF4-induced/plasma cell signatures (Figure 1B). Consistent with increased mitochondrial activity in VL51-Ibru, the latter showed enrichment of genes in pathways related to mitochondrial activation and downregulation of gene sets associated with mitochondrial repression (Figure 1B).

Given the upregulation of secreted factors observed at the RNA level, we tested the sensitivity of VL51 PAR cultured in resistant-conditioned medium to ibrutinib. Consistent with the role of secreted factors in resistance, we observed decreased sensitivity to ibrutinib in PAR cultured for 48 hours in VL51-Ibru-conditioned medium (supplemental Figure 5). Subsequently, we investigated the secretome profile of conditioned media

(72 hours) from PAR and VL51-Ibru using a cytokine array, which confirmed the secretion of IL-16 and CXCL10 (Figure 1A; supplemental Figure 6; supplemental Table 4). In line with the in vitro secretome and RNA-seq findings, VL51-Ibru exhibited higher intracellular IL-16 expression than PAR (supplemental Figure 7).

In immunoblotting, we observed increased levels of phosphorylated AKT (pAKT-S473) and BTK (pBTK-Y223) and the upregulation of total and phosphorylated SYK (pSYK-Y352) in VL51-Ibru cells compared with their parental counterpart (Figure 1C; supplemental Figure 8). We did not detect any pSYK-YY525/526 in VL51 parental or VL51-Ibru cells (Figure 1C; supplemental Figure 8).

IL-16 induces resistance to ibrutinib in lymphoma cells

Subsequently, we investigated whether IL-16 stimulation could reduce the response to ibrutinib. IL-16 signaling depends on binding to 2 potential surface receptors: CD4 or CD9.¹⁵ Because the T-cell marker CD4 is not expressed in B cells, as confirmed at the RNA level in our models, we examined CD9 surface expression in VL51 parental and resistant cells. CD9 was well expressed in both, with slightly higher expression in the resistant cells (supplemental Figure 9). Recombinant IL-16 induced resistance to ibrutinib in VL51 PAR in a dose-dependent manner, with a limited effect in resistant cells, consistent with the constitutive secretion of IL-16 in the latter (Figure 2A-B; supplemental Figure 10). To further investigate the impact of IL-16 on the response to ibrutinib, we stimulated 2 CD9⁺ B-cell lymphoma models, selected based on previous in-house transcriptomic data,¹⁶ with recombinant IL-16. We observed reduced ibrutinib sensitivity in REC1 (MCL), SSK41 (MZL), and MEC1 (CLL) cells (Figure 2C; supplemental Figure 11).

IL-16/CD9 axis is expressed in B-cell lymphoid neoplasms, and IL-16 is enriched in the serum of patients with CLL who do not respond to ibrutinib

Subsequently, we investigated the potential relevance of the IL-16/CD9 axis in B-cell lymphoma clinical specimens using public expression data sets.¹⁷⁻²⁰ IL-16 and CD9 were expressed across B-cell lymphoma subtypes, and CLL cells express surface CD9 (coexpressed with CD19). In patients with CLL treated with ibrutinib,²¹ IL-16 expression increased during treatment. We then measured IL-16 levels in the serum of patients with CLL with resistance to the BTK inhibitor in the absence of *BTK* or *PLCG2* mutations, compared with patients responding to the drug and matched for clinical features (supplemental Table 5). In agreement with our hypothesis, resistant patients exhibited significantly higher levels of IL-16 than responders (Figure 3).

IL-16-triggered transcriptome and proteome changes and pathway activation

Consequently, we investigated transcriptomic and proteomic changes orchestrated by IL-16 in VL51 PAR. VL51 PAR were stimulated with 30 ng/mL for 24 hours and underwent RNA-seq and proteomic profiling. Upon exposure to IL-16, VL51 PAR exhibited enrichment in proliferation signatures (MYC and E2F targets and G2M checkpoint) and PI3K/AKT/mTOR and MAPK

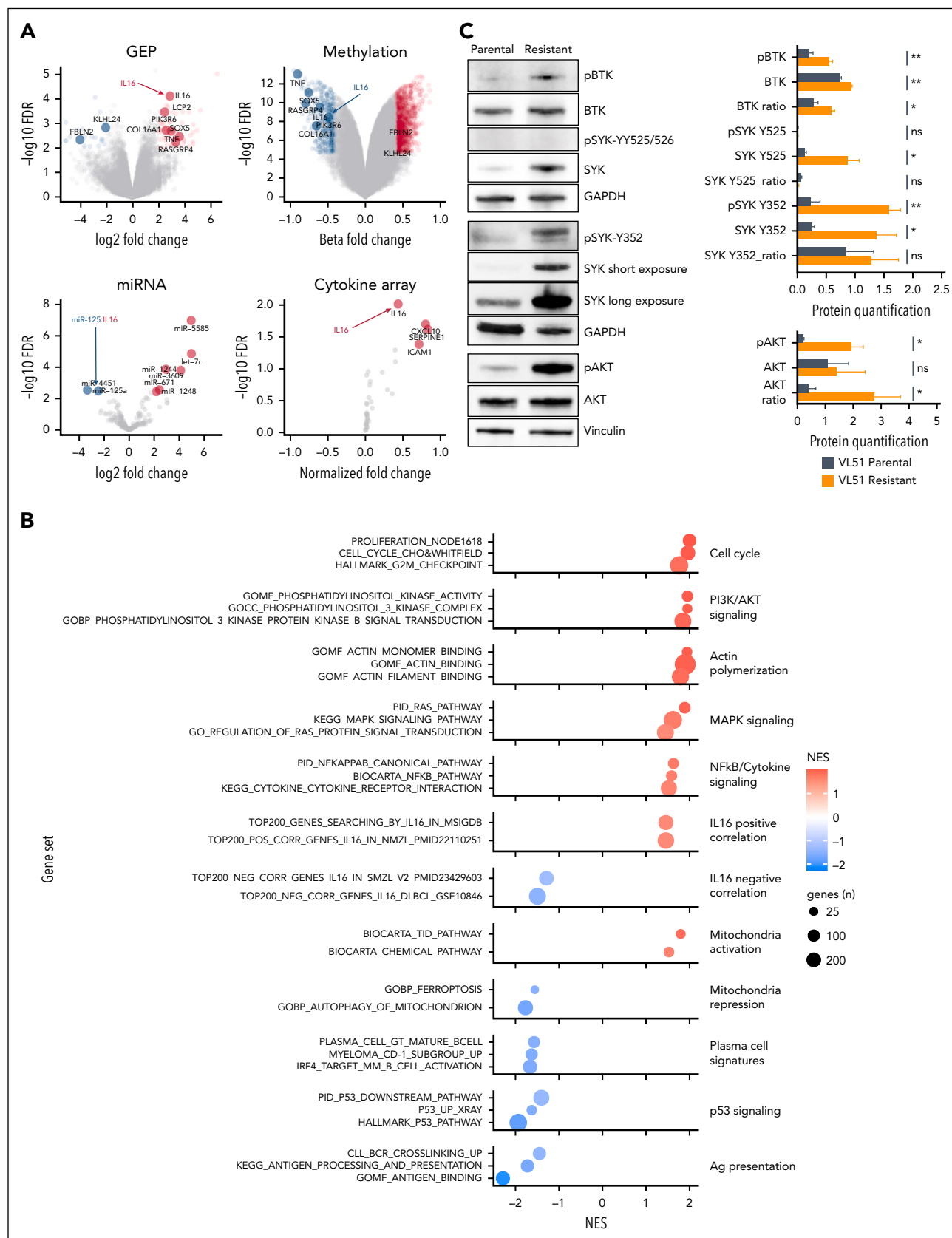


Figure 1. A multiomics signature of resistance identifies IL-16 overexpression and secretion. (A) Volcano plot of RNA (GEP), methylation, miRNA, and secretome (cytokine array) profiles of resistant cells compared with PAR. Volcano plots show the fold change between resistant cells and PAR in the x-axis ($\log_2$ fold change for RNA and miRNA, β -value fold change for methylation, and normalized fold change for cytokine array) and $-\log_{10}$ FDR (Benjamin-Hochberg correction) on the y-axis. Values from a moderated *t* test. Genes upregulated and hypomethylated are highlighted in red for the resistant cells and in blue for the parental line. (B) Top enriched gene sets in

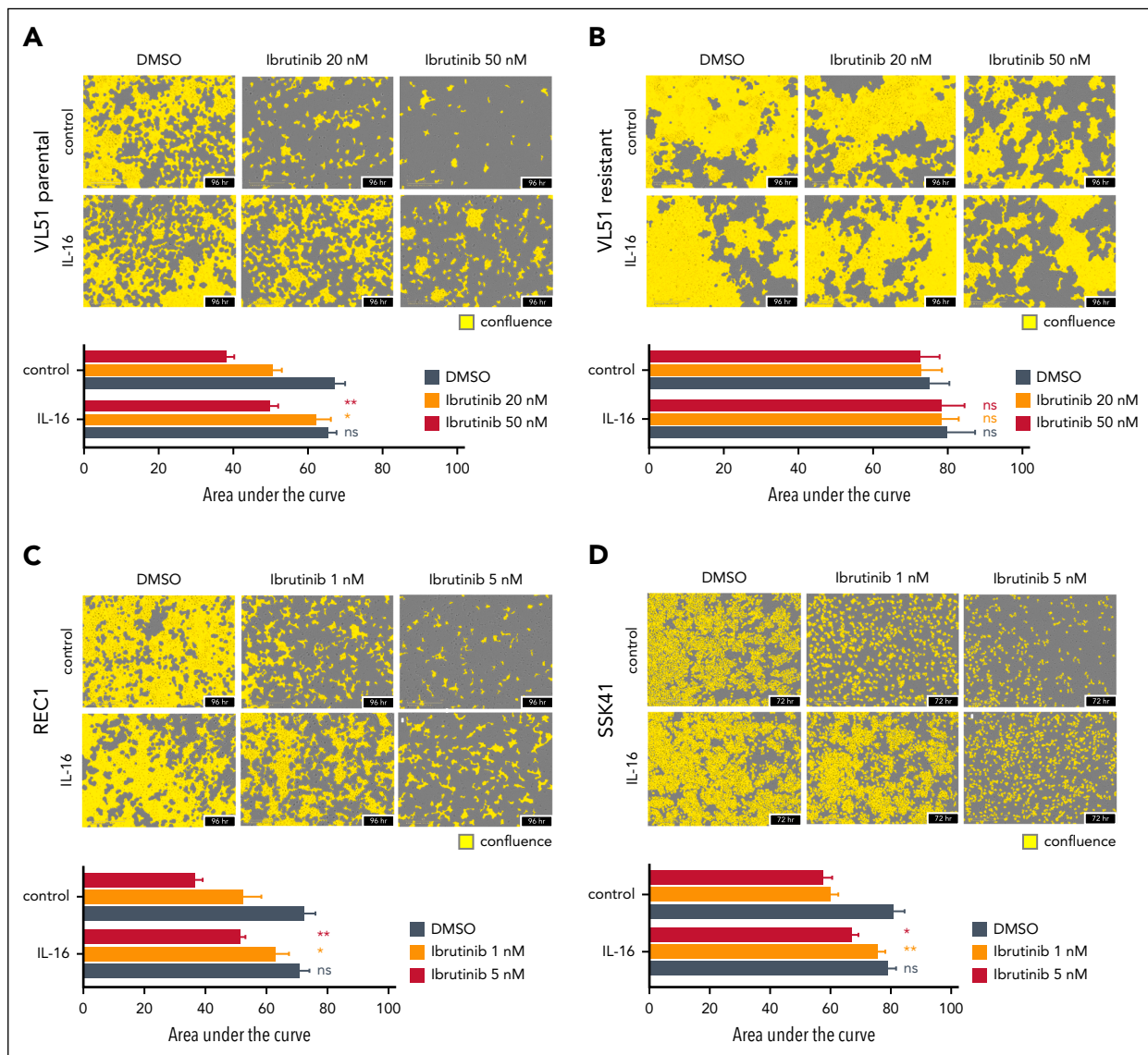


Figure 2. Recombinant IL-16 induces resistance to ibrutinib in B-cell lymphoma models. VL51 parental (A), ibrutinib-resistant VL51 (B), REC1 (C), and SSK41 (D) cells were cultured for 6 days in the presence of recombinant human IL-16 (30 ng/mL) or phosphate-buffered saline (PBS; control) and then exposed to increasing concentrations of ibrutinib or dimethyl sulfoxide (DMSO; control). Cell growth was monitored longitudinally using the IncuCyte S3 live-cell imaging system, and proliferation was quantified as percent area confluence (yellow mask). Representative IncuCyte images at the indicated time point (black box) are shown. Bar graphs depict the area under the growth curves (AUC; mean $\pm$ SD) for quantitative comparison between conditions. Data represent at least 2 independent experiments. Statistical significance was assessed using a Z-test comparing IL-16-treated vs control (PBS) conditions. * $P < .05$; ** $P < .01$; *** $P < .001$.

signaling pathways, while downmodulating oxidative phosphorylation (OXPHOS) and apoptosis (supplemental Table 6).

Among the overexpressed proteins, there was a marked enrichment of components of the PI3K/AKT/mTOR axis. These included upstream activators (TEC, FYN, LRP8, ITGB7, IL12RB1, and P2RY8), core signaling mediators (PIK3R6, SESN2, and AURKA), and AKT-driven cell cycle regulators (PLK1, CCNB1, CCNB2, CCNA2, E2F7, and BUB1B). Consistent with mTOR pathway activation, we also observed an

increased expression of metabolic effectors, including glutaminase, stearyl-CoA desaturase, multiple solute carrier transporters, and phospholipid transfer protein, as well as AKT-regulated epigenetic modulators (DNMT1 and PHF19).

In addition, the upregulated proteome encompassed key components of the RAS/MAPK cascade (RET, TRIB1, and MEF2B), transcriptional regulators (STAT4), and mediators of BCR- and NF- κ B-dependent signaling (CCL3, SYK, and TNFRSF13B). The levels of several antiapoptotic proteins were

Figure 1 (continued) resistant cells (red) and PAR (blue) by gene set enrichment analysis (GSEA). (C) Protein levels of total and phosphorylated BTK, SYK, and AKT by immunoblot in parental (gray) and resistant (orange) cells. Values correspond to the average of protein quantification normalized to glyceraldehyde-3-phosphate dehydrogenase (GAPDH) or vinculin levels in 2 independent experiments (the second experiment in supplemental Figure 8). The error bars represent the standard deviation (SD) of the mean. Statistical significance was assessed using a nonparametric t test comparing resistant vs parental. * $P < .05$; ** $P < .01$. FDR, false discovery rate; GEP, gene expression profile (RNA-sequencing); NES, normalized enrichment scores; ns, not significant.

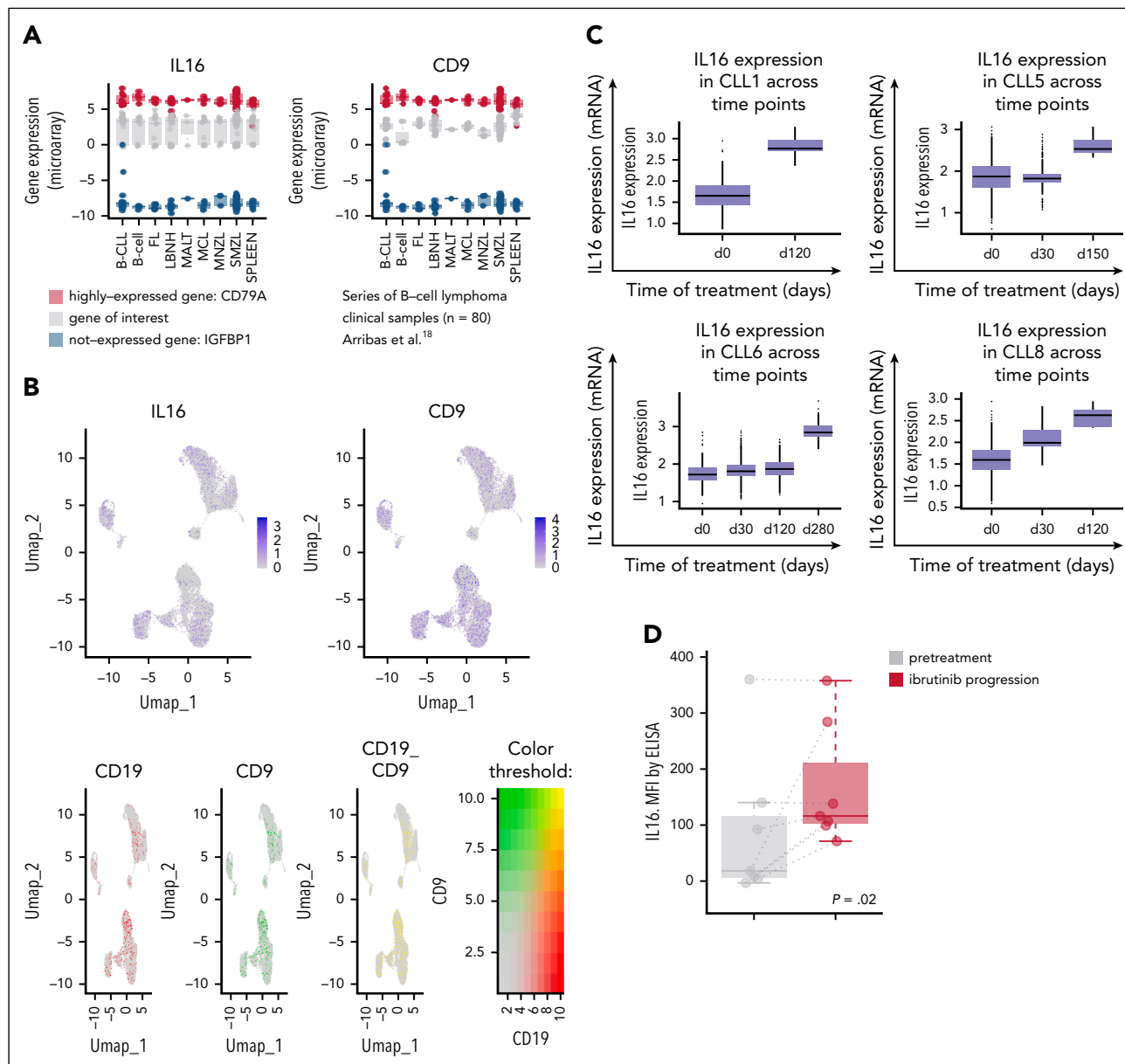


Figure 3. IL-16 expression is increased in CD9⁺ malignant B cells and rises at disease progression in CLL. The potential relevance of the IL-16/CD9 axis in B-cell lymphoma was investigated in clinical specimens using public expression data sets.^{18,21} (A) Expression levels of genes related to ibrutinib/R-CHOP resistance were studied across different subtypes of B-cell lymphoma (n = 80).¹⁸ Red indicates highly expressed genes in B cells (CD79A), whereas Blue indicates genes not expressed in B cells (IGFBP1) and black indicates the gene of interest. The expression of IL-16, CD9, and CD19 was evaluated in publicly available single-cell RNA-seq data sets.²¹ (B) UMAP projections of single-cell transcriptomic data illustrating the distribution of IL-16 and CD9 expression (top), CD19 and CD9 expression (bottom left and middle), and coexpression of CD19 and CD9 (bottom right). Color intensity reflects normalized gene expression levels, as indicated. (C) Longitudinal analysis of IL-16 mRNA expression in purified CLL cells from individual patients across clinical time points (pretreatment and progression on therapy). The box plots represent the median, interquartile range, and whiskers, and the dots indicate individual samples. (D) Serum IL-16 protein levels measured by immunoassay (Luminex; R&D Systems) in paired samples from patients with CLL obtained before treatment (gray) and at disease progression (red). Each pair of connected dots represents an individual patient. IL-16 levels are significantly increased at progression (paired analysis; $P = .02$). B-CLL, B-cell CLL; ELISA, enzyme-linked immunosorbent assay; FL, follicular lymphoma; LBNH: unspecified non-Hodgkin B-cell lymphoma; MALT, MZL of the mucosa-associated tissue, MFI, mean fluorescence intensity; NMZL: nodal MZL; SMZL: splenic MZL; UMAP, uniform manifold approximation and projection.

also increased, including BCL2A1 (BFL1), TNFRSF17 (BCMA), and TNFRSF1B (TNFR2), highlighting the activation of convergent survival programs (supplemental Figure 12; supplemental Table 6).

Consistent with the role of IL-16 in resistance to ibrutinib, the gene expression profile of VL51-lbru showed enrichment of the recombinant IL-16 signature and recombinant IL-16 induced the ibrutinib-resistant signature, including upregulated genes

related to resistance to targeted therapies (IGF1, EGR1/2, and BCL2A1) (supplemental Figure 13). IL-16 stimulation repressed genes involved in the negative regulation of actin polymerization, consistent with the observed enrichment of actin polymerization signatures in VL51-lbru cells (supplemental Figure 14). Notably, genes involved in mitochondrial biogenesis were upregulated in PAR upon recombinant IL-16, whereas apoptosis was reduced by exposure to the cytokine (supplemental Figure 15).

Based on the integrated transcriptome and proteome orchestrated by IL-16 in PAR, we investigated signaling activation in the VL5 cell lines. Consistent with the multiomics data, recombinant IL-16 increased levels of pBTK-Y223, pAKT-S473, and pSYK-Y352 in VL51 PAR, resembling the signaling cascade observed in VL51-lbru. Despite the upregulation of SYK in the proteomic profiles, immunoblotting showed only a limited effect of recombinant IL-16 on the total SYK levels (supplemental Figure 16).

Given that CXCL10 is an NF- κ B target, we wondered whether IL-16 might induce CXCL10 secretion via NF- κ B activation in the ibrutinib-resistant derivative cells. Using real-time polymerase chain reaction, we measured CXCL10 expression in VL51 cells after exposure to recombinant IL-16 and observed IL-16-induced upregulation of CXCL10. Supporting a potential role for CXCL10 in ibrutinib resistance, its receptor CXCR3 was well expressed in VL51 cells (surface expression by fluorescence-activated cell sorting). Therefore, we tested the effect of recombinant CXCL10 on the response to ibrutinib in VL51 PAR and observed a decrease in the ibrutinib response upon CXCL10 stimulation (supplemental Figure 17).

Finally, we investigated whether these results could be extrapolated to other *in vitro* models. In a panel of 13 B-cell lymphoma cell lines that we had previously treated with the BTK inhibitor (supplemental Figure 18), IL-16 and CD9 expression levels, but not CXCL10, were inversely correlated with ibrutinib sensitivity.

IL-16/CD9 axis is expressed in DLBCL clinical specimens and confers R-CHOP resistance in ABC DLBCL cells

Using public expression data sets,¹⁷⁻²⁰ we observed that both IL-16 and CD9 were expressed across clinical specimens of non-Hodgkin lymphoma and in malignant cells of DLBCL specimens (supplemental Figure 19).

Considering that IL-16 is part of the ABC DLBCL transcriptome signature,²²⁻²⁴ we tested the effect of IL-16 stimulation on R-CHOP (rituximab, cyclophosphamide, hydroxydaunorubicin, vincristine [Oncovin], and prednisone) sensitivity in ABC DLBCL cell lines. OCI-Ly10 and TMD8 cell lines showed high sensitivity to R-CHOP *in vitro* and were tested for changes in drug response upon stimulation with recombinant IL-16. We observed that IL-16 stimulation decreased sensitivity to R-CHOP in ABC DLBCL cell lines (supplemental Figure 20), whereas cell proliferation was not affected by the cytokine.

Pharmacological inhibition of the IL-16/CD9/PI3K axis improves the sensitivity of MZL and MCL cells to BTK inhibitors and R-CHOP

We then explored blocking the IL-16/CD9 axis using an anti-IL-16 neutralizing antibody²⁵ or an 8-mer peptide that binds and inhibits CD9 (CD9-binding peptide [CD9-BP]).²⁵ Therefore, upon IL-16 stimulation, we exposed PAR and VL51-lbru cells, as well as other CD9⁺, ibrutinib-sensitive cell lines, to ibrutinib alone and in combination with IL-16/CD9-targeting compounds. Combining either the anti-IL-16 antibody or CD9-BP with ibrutinib restored ibrutinib sensitivity in VL51-lbru cells and reverted IL-16-driven resistance in PAR (Figure 4A-B).

Similar to VL51 cells, the combination of ibrutinib with the anti-IL-16 antibody or CD9-BP was beneficial in additional MZL and MCL models *in vitro* (SSK41 and REC1, respectively; Figure 4B). Notably, in the IL-16-positive, ibrutinib-resistant MAVER-1 (MCL) model, the addition of the anti-IL-16 antibody was synergistic, enhancing the response to ibrutinib, and IL-16 stimulation conferred resistance to zanubrutinib in MZL and MCL models. As negative controls, we tested the CD9-low Karpas1718 and its ibrutinib-resistant derivative (Karpas1718-lbru).⁷ No effect was observed upon IL-16 stimulation or CD9 blockade (supplemental Figure 21A-C).

We then assessed the role of the IL-16/CD9 axis in the response to R-CHOP in 4 ABC DLBCL cell lines expressing IL-16 and CD9 (OCI-Ly10, TMD8, U2932, and RIVA). The addition of the blocking anti-IL-16 antibody or CD9-BP enhanced R-CHOP sensitivity in all the models (Figure 4C-E).

Given the IL-16-induced activation of AKT that we observed, we next investigated the combination of ibrutinib with the PI3K δ inhibitor idelalisib under IL-16 stimulation. As with IL-16/CD9 interference, idelalisib-mediated inhibition of the PI3K/AKT cascade decreased the IL-16-induced increase in pAKT, improved ibrutinib sensitivity in the VL51 cells and in the SSK41 and REC1 models, and reversed IL-16-driven resistance in the ABC DLBCL cell lines (Figure 4A-C; supplemental Figure 21D).

As a mechanism of action, IL-16 reversed the ibrutinib-induced G0 to G1 cell cycle arrest in PAR and the addition of CD9-BP restored ibrutinib's cytostatic effect. In VL51-lbru cells, ibrutinib exposure alone did not affect the cell cycle, but its combination with either CD9-BP or the anti-IL-16 antibody arrested the resistant cells (supplemental Figure 22).

Collectively, these data suggest a role for IL-16 in resistance not only to BTK inhibitors in MZL and MCL but also to the first-line chemoimmunotherapy in ABC DLBCL. These findings support the therapeutic potential of targeting the IL-16/CD9 axis to overcome drug resistance and improve treatment efficacy in this aggressive lymphoma subtype.

CD9 blocking abrogates IL-16-driven prosurvival signaling and increases the activity of BTK inhibitors in cell lines

To elucidate the mechanism of action of the CD9-BP, we performed a proteomic analysis of VL51 PAR exposed to the CD9-BP alone or in combination with IL-16, alongside phosphate-buffered saline as a negative control, and with IL-16 (as already shown in Figure 4). Consistent with the lack of effect on cell viability, the CD9-BP, as a single agent, modulated only a limited number of proteins, whereas, when added to recombinant IL-16, it markedly altered the changes induced by interleukin. The IL-16-driven enrichment of proliferation signatures (MYC targets, E2F, and G2M checkpoint) and signaling cascades (PI3K/AKT/mTOR and RAS/MAPK), together with the repression of apoptosis and OXPHOS, was fully abrogated by the addition of CD9-BP (Figure 5; supplemental Figure 23).

Mechanistically, consistent with the signaling observed in VL51-lbru, recombinant IL-16 induced AKT, BTK, and SYK activation in VL51 PAR, an effect blocked by the addition of

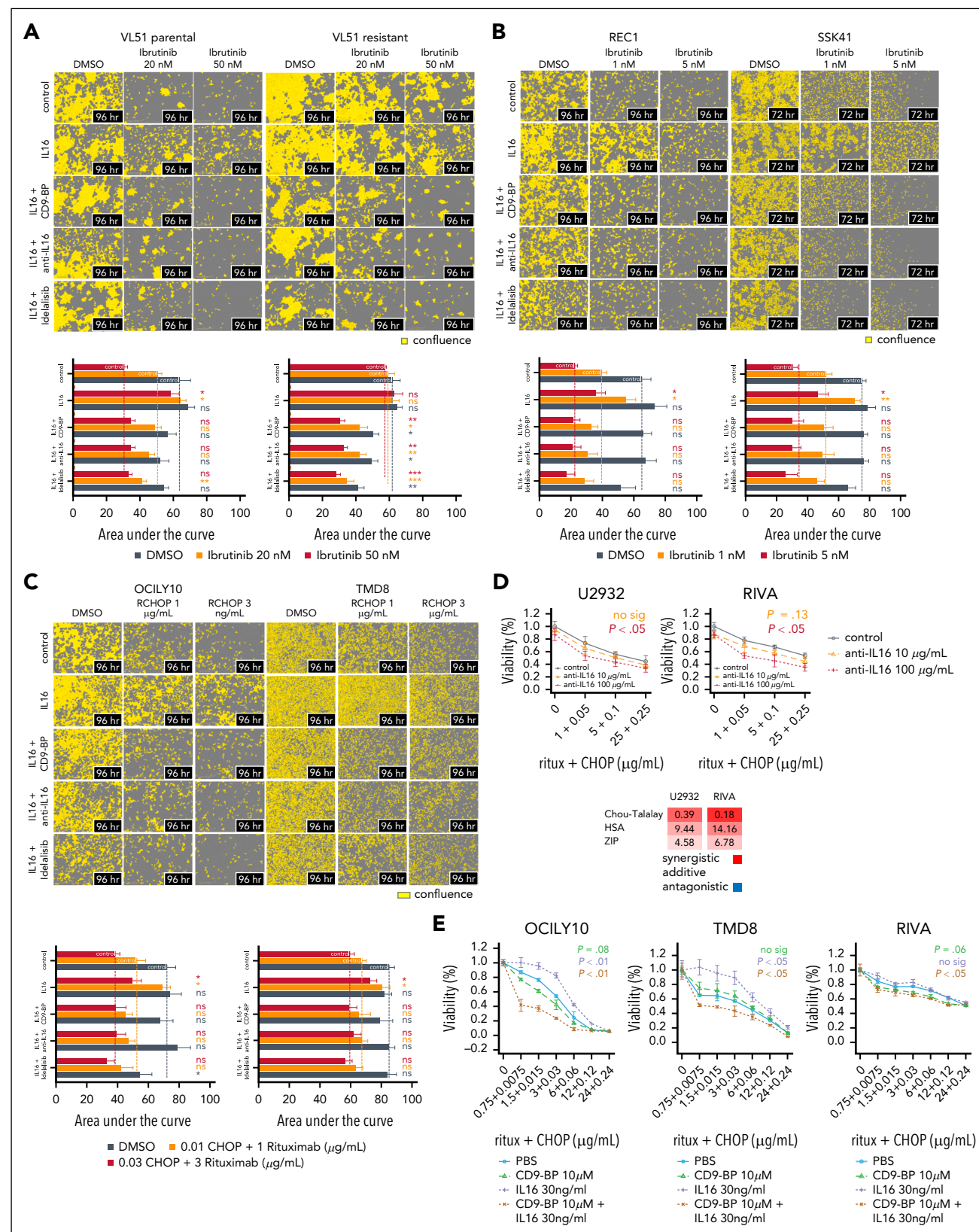


Figure 4. Blocking the IL-16/CD9/PI3K axis improves response to ibrutinib and R-CHOP in B-cell lymphoma. Response to ibrutinib was evaluated in VL51 parental and ibrutinib-resistant models (A) and REC1 and SSK41 models. (B) R-CHOP sensitivity was evaluated in OCI-Ly10 and TMD8, U2932, and RIVA (C-E). Panels A-C show representative IncuCyte S3 live-cell images of lymphoma cells treated with vehicle (DMSO) or increasing concentrations of ibrutinib, combined with isotype antibody (IgG2 at 10 µg/mL, control), IL-16 (30 ng/mL), CD9-BP (1 µM), the blocking antibody anti-IL-16 (10 µg/mL), or idelalisib (20 nM). The yellow masks indicate cell confluence. Images were acquired at the indicated time point (black box) and are representative of at least 2 independent experiments. Quantification of cell growth expressed as area under the confluence-time curve (AUC) over the treatment period. For each condition, the bars are ordered top to bottom as control (isotype antibody in PBS), IL-16 stimulation

CD9-BP. In VL51-Ibru, exposure to CD9-BP reduced pBTK, pAKT, and pSYK-Y352 (Figure 5C; supplemental Figure 23C). BFL1 and pSYK-YY525/526 levels were not detectable (Figure 5C; supplemental Figure 23C).

We also silenced CD9 with small interfering RNA in VL51 and additional MCL and ABC DLBCL models (REC1 and TMD8, respectively; supplemental Figure 24). CD9 knockout blocked IL-16–induced pAKT and BFL1 increases and reverted IL-16–driven ibrutinib resistance in VL51 PAR, REC1, and TMD8 and the acquired resistance in VL51-Ibru (Figure 5D–G).

CD9 blocking abrogates IL-16–driven prosurvival signaling and increases the activity of BTK inhibitors in CLL primary cells

To further assess the clinical relevance of our findings, we investigated the effects of IL-16 and CD9-BP on signaling pathways in primary CLL samples. CD19⁺-enriched primary cells from 3 CLL donors (supplemental Figure 25) were exposed to IL-16, CD9-BP, the combination of both, or phosphate-buffered saline as a control. First, in primary CLL cells, IL-16 reduced ibrutinib sensitivity, which was restored by the addition of CD9-BP. Second, based on the enrichment of signaling cascades observed in the proteomic profile of VL51 cells, the levels of pAKT-S473 and phosphorylated extracellular signal-regulated kinase (pERK; pERK-T202/Y204) were evaluated by intracellular staining. In line with a role for the IL-16/CD9 signaling axis in activating signaling cascades, we observed increased phosphorylation of AKT and ERK upon IL-16 stimulation, which was abrogated by CD9-BP. Finally, the IL-16–induced upregulation of antiapoptotic effectors was confirmed in primary CLL cells, in which CD9⁺ specimens showed increased BFL1 and RELB (Figure 6). We observed a positive trend between IL-16–induced increased phosphorylation and CD9 positivity (supplemental Figure 24D).

Chromatin remodeling facilitates IL-16 transcription through FLI1 transcription factor

Finally, we analyzed the chromatin accessibility of the VL51-Ibru cells to elucidate the epigenetic reprogramming underlying the resistance mechanism. The analysis revealed significant changes compared with the PAR line. Specifically, we observed increased chromatin accessibility in regions associated with genes upregulated at the RNA level (supplemental Figure 26A). In contrast, regions with decreased chromatin accessibility corresponded to downregulated genes in the RNA-seq data (supplemental Figure 26B).

Both the promoter and intergenic regions of the IL-16 gene were consistently among the most accessible regions in resistant cells, and the latter showed enrichment of peaks at the IL-16 promoter compared with PAR (Figure 7A–B). We then performed region set enrichment analysis to investigate the functional implications of epigenetic reprogramming. This analysis revealed increased activity of transcription factors

involved in B-cell development and function, including EBF1, PAX5, PU.1, and ETS family members such as FLI1 (Figure 7C). Notably, FLI1 is known to induce the expression of genes involved in immune response, interleukin signaling, and BCR signaling.²⁶ Indeed, FLI1 and its transcriptional signature were upregulated at the RNA level in the ibrutinib-resistant models (supplemental Figure 26C–D). In addition, IL-16 was identified as a directly FLI1-upregulated and clinically correlated target in the germinal center B-cell-like DLBCL subtype (supplemental Figure 26E).²⁶

To assess the relationship between FLI1 and IL-16, we first validated the binding of FLI1 at the IL-16 promoter region by chromatin immunoprecipitation quantitative polymerase chain reaction in a germinal center B-cell-like lymphoma cell line (supplemental Figure 26F) and then performed a stable knockdown of FLI1 in the ibrutinib-resistant cell line VL51 using short hairpin RNA, observing the downregulation of the interleukin (Figure 7D). Finally, the ibrutinib-resistant cells were more sensitive than the PAR to chelerythrine, an antitumor agent that inhibits FLI1,²⁷ and the combination with ibrutinib was beneficial in both parental and resistant cells, with an enhanced synergistic effect in the latter (supplemental Figure 26G–H). All these data support the role of FLI1 in the IL-16–driven resistance to ibrutinib in lymphoma cells.

Discussion

In this study, we developed and characterized a novel model of secondary resistance to BTK-targeting agents in splenic MZL, identifying the IL-16/CD9 axis as a previously unrecognized mechanism of resistance to targeted agents and chemotherapy in B-cell lymphomas, which can be pharmacologically inhibited.

Resistance to BTK inhibitors remains a major therapeutic challenge in B-cell malignancies. We established an ibrutinib-resistant MZL cell line (VL51-Ibru) displaying a 15-fold increase in IC₅₀ compared with parental VL51 cells. Resistance was stable after drug withdrawal, indicating a durable adaptive state rather than transient drug tolerance. VL51-Ibru cells were resistant to covalent and noncovalent BTK inhibitors and to BTK degraders.

Genomic profiling did not reveal mutations in *BTK* or *PLCG2*, 2 genes most frequently implicated in clinical ibrutinib resistance.^{1,3,28} Instead, transcriptomic, proteomic, and epigenetic analyses revealed extensive reprogramming, with the upregulation of genes involved in BTK activation, cell proliferation, and PI3K-AKT and MAPK signaling. IL-16 emerged as the most highly upregulated cytokine at both RNA and protein levels, accompanied by the downregulation of its inhibitory miRNAs miR-125a and miR-4451.¹⁴ Conditioned media from resistant cells conferred ibrutinib resistance to parental VL51 cells, supporting a paracrine mechanism. Recombinant IL-16 was sufficient to induce resistance in parental VL51 cells and in

Figure 4 (continued) (30 ng/mL), and IL-16 combined with either CD9-BP (1 μM), a neutralizing anti-IL-16 antibody (10 μg/mL), or idelalisib (20 nM), as indicated. The bars represent mean ± SD. The dashed vertical lines indicate the mean AUC of isotype-treated controls. (D–E) U2932 and RIVA cells were cultured for 72 hours in the presence of the blocking antibody for IL-16 (10 μg/mL in yellow or 100 μg/mL in red) or PBS (control, black) under exposure to increasing concentrations of the *in vitro* equivalent of R-CHOP, followed by MTT assay. Three independent experiments were performed. Statistical comparisons were performed using a Z-test between all conditions and the corresponding control. **P* < .05; ***P* < .01; ****P* < .001. IgG2, immunoglobulin G2.

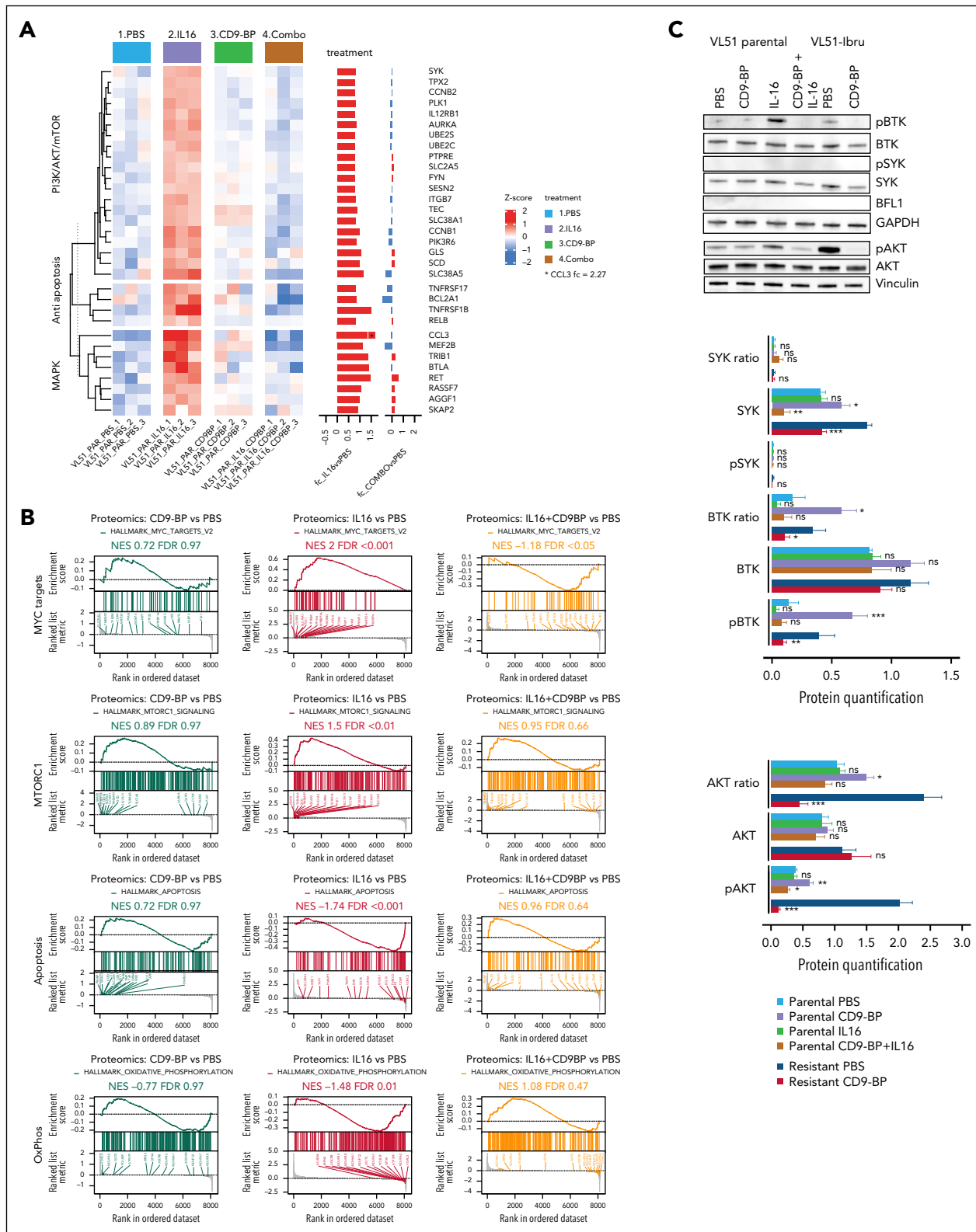


Figure 5. Modulation of IL-16-associated signaling programs by CD9 blockade in lymphoma cells. (A) Heat map of selected differentially expressed proteins from quantitative proteomic analysis of VL51 PAR treated with PBS, recombinant IL-16 (30 ng/mL), CD9-BP (10 μ M), or the combination of IL-16 and CD9-BP. Protein abundance is shown as z scores, and proteins are organized according to functional categories, including PI3K/AKT/mTOR signaling, MAPK signaling, and antiapoptotic pathways, as indicated. (B) GSEA based on proteomic data comparing CD9-BP vs PBS, IL-16 vs PBS, and IL-16 plus CD9-BP vs PBS. Enrichment plots are shown for representative pathways, illustrating changes in NES across treatment conditions. (C) Representative immunoblot analysis of SYK, BTK, AKT, and their phosphorylated forms in VL51 parental and ibrutinib-resistant cells treated with PBS, IL-16, CD9-BP, or the combination, as indicated. The bar graphs show densitometric quantification normalized to total protein levels and expressed relative to PBS controls. Data represent mean $\pm$ SD of protein quantification normalized to vinculin/GAPDH from at least 2 independent experiments. Statistical comparisons between conditions are indicated. * $P < .05$; ** $P < .01$; *** $P < .001$. (D-G) Effects of CD9 silencing (small interfering RNA [siRNA]) on

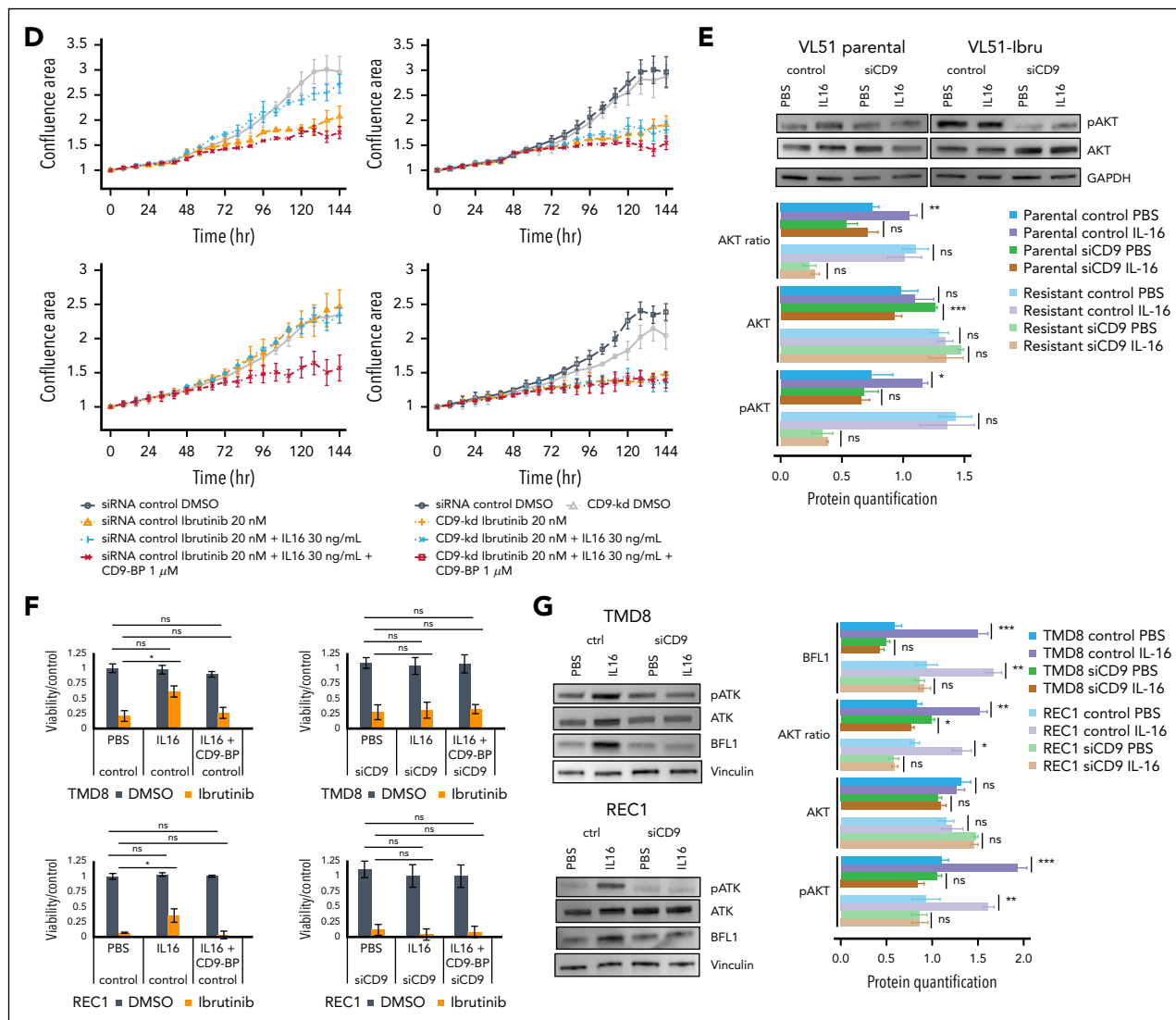


Figure 5 (continued) IL-16-associated signaling and drug response in B-cell lymphoma models. (D) InCuCyte S3 live-cell growth analysis of VL51 parental (top) and ibrutinib-resistant (bottom panel) cells transfected with control siRNA or CD9-targeting siRNA (siCD9) and treated with vehicle (DMSO) or ibrutinib, in the presence of recombinant IL-16 (30 ng/mL) or PBS (control). Cell confluence was monitored longitudinally and is expressed as the fold increase from time zero. Data represent the mean $\pm$ SD of at least 2 independent experiments. (E) Representative immunoblot analysis and densitometric quantification of AKT and pAKT in VL51 parental and ibrutinib-resistant cells transfected with control siRNA or siCD9 and treated with PBS or IL-16 (30 ng/mL), as indicated. Protein levels were normalized to loading controls. (F-G) Quantification of cell viability and signaling outputs in TMD8 and REC1 cells following CD9 silencing (siCD9), assessed by immunoblot densitometry and viability assays (MTT) under PBS or IL-16 (30 ng/mL) stimulation and with or without ibrutinib (1nM) treatment. The bar graphs represent mean $\pm$ SD from independent experiments. For all panels, statistical comparisons between conditions are indicated. * P < .05; ** P < .005; *** P < .0005. Combo, IL-16 plus CD9-BP; Ctrl, control.

additional CD9⁺ models derived from MZL, MCL, and CLL. These findings suggest that IL-16 contributes to a protective microenvironment that limits the efficacy of BTK inhibitors. Consistent with this, B-cell lymphoma clinical specimens expressed IL-16 and CD9, and serum IL-16 levels were elevated in patients with ibrutinib-resistant CLL lacking *BTK* or *PLCG2* mutations, implicating IL-16 as an alternative resistance pathway.

IL-16 has been associated with inflammatory disorders,²⁹ cutaneous T-cell lymphoma,³⁰ and high disease burden in CLL and multiple myeloma.^{15,31-33} IL-16 blockade reduces myeloma cell proliferation,^{15,32} and IL-16 neutralization suppresses tumor growth in murine colorectal cancer models.³⁴

Our data and the literature indicate a complex cytokine-driven resistance network in B-cell malignancies, in which IL-16 contributes to cell-intrinsic survival and a broader immunomodulatory microenvironment that can promote resistance to targeted therapies such as ibrutinib.

We demonstrated that IL-16 signals through the noncanonical receptor CD9 rather than CD4.¹⁵ CD9 is a tetraspanin that regulates adhesion, migration, and immune signaling, induces PI3K and AKT phosphorylation, and modulates immune responses.³⁵⁻³⁸ In B cells, CD9 colocalizes with CD19 and the BCR complex and promotes tyrosine phosphorylation.^{39,40} CD9 is expressed in ABC DLBCL, Waldenström macroglobulinemia, and multiple myeloma, in which high expression

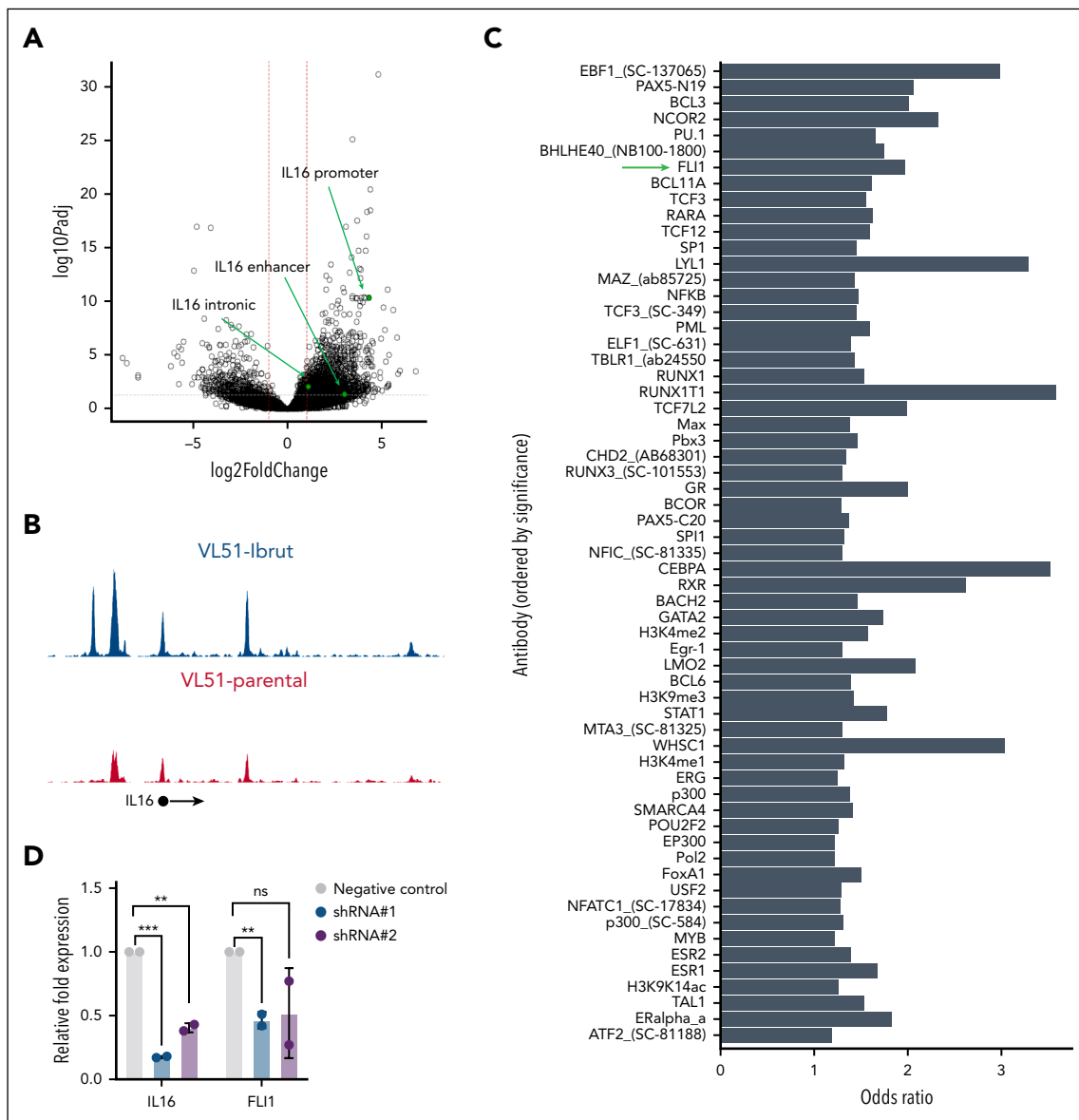


Figure 7. Chromatin remodeling facilitates IL-16 transcription through the FLI1 transcription factor. (A) Volcano plots showing differential chromatin accessibility across the set of consensus peaks from ATAC-seq data. The dashed lines indicate significance thresholds (absolute $\log_2$ fold change >1 ; adjusted $P < .05$). Highlighted are differentially accessible peaks at the IL16 locus in the VL51 ibrutinib-resistant cell line compared with its parental counterpart. (B) Chromatin accessibility tracks around the IL16 locus for VL51 ibrutinib-resistant cells (top) and PAR (bottom). Enrichment is visualized as reads per bin per million base pairs. The black arrow indicates the transcription start site of the IL16 gene. (C) Region set enrichment analysis of differentially accessible regions in the VL51 ibrutinib-resistant model, performed using LOLA software and based on publicly available ChIP-seq data (ENCODE and CODEX). Results are ranked by significance (q value). (D) Relative mRNA expression of IL-16 and FLI1 (normalized to GAPDH) upon FLI1 silencing with 2 different short hairpin RNAs (shRNAs) in the VL51 ibrutinib-resistant model, compared with the negative shRNA control. Asterisk (*) indicates $P < .05$ from a t test comparing each shRNA vs negative control. ATAC-seq, assay for transposase-accessible chromatin sequencing; ChIP-seq, chromatin immunoprecipitation sequencing.

increased expression of the antiapoptotic effector BCL1 (BCL2A1), a known mediator of resistance to venetoclax and R-CHOP in DLBCL and CLL,⁴³⁻⁴⁵ providing a direct mechanistic link between IL-16 signaling and therapeutic resistance.

Within this framework, SYK emerges as a critical signaling node downstream of the IL-16/CD9 axis. Both total SYK and its activated phosphorylated form, pSYK-Y352, were increased in VL51-Ibru cells and were induced by IL-16 exposure. Phosphorylation at Y352 is sufficient to activate downstream AKT and ERK signaling,⁴⁶ whereas elevated total SYK levels may

also contribute through kinase-independent scaffold functions that stabilize CD9-associated signaling complexes. In this manner, SYK integrates PI3K/AKT and ERK pathway activation, sustaining prosurvival signaling even in the setting of BTK inhibition. The ability of CD9 blockade to abolish these effects further underscores the functional dependence of this circuitry on the IL-16/CD9 axis.

Epigenetic analyses indicated that IL-16 upregulation in resistant cells is driven by increased FLI1 activity. Chromatin accessibility was increased at the IL-16 locus and FLI1

knockdown reduced IL-16 expression, implicating adaptive chromatin remodeling in resistance.²⁶

Morphologically, VL51-Ibru cells exhibited increased membrane protrusions and mitochondrial remodeling, features associated with enhanced signaling and metabolic flexibility. Similar cytoskeletal and mitochondrial adaptations have been linked to BTK inhibitor resistance.^{6,47-49} IL-16/CD9 signaling has been shown to regulate actin dynamics and migration,⁵⁰ and recombinant IL-16 induces actin polymerization and mitochondrial biogenesis programs.⁵¹

These findings place IL-16 within a broader network of cytokine-driven resistance mechanisms, alongside CCL3, CCL4, IL-6, and HB-EGF.^{7,9,52-56} Clinically, IL-16 may serve as a biomarker of resistance in patients lacking *BTK* or *PLCG2* mutations. Integration with circulating tumor DNA analysis⁵⁷ may enable the earlier detection of resistance. Collectively, our data support the clinical evaluation of IL-16- or CD9-directed therapies, alone or in rational combinations with BTK inhibitors or standard chemoimmunotherapy. Future directions should focus on prospectively validating these findings in patient samples and assessing whether IL-16 inhibition can enhance BTK inhibitor efficacy in clinical settings. Given the role of IL-16 in modulating the tumor microenvironment, additional studies should examine its effects on immune cell interactions and whether IL-16 blockade can enhance responses to immune-based therapies.

In conclusion, our study identifies the IL-16/CD9 axis as a novel resistance mechanism to BTK inhibitors and R-CHOP in B-cell lymphomas. Targeting this axis may offer a promising strategy to overcome resistance and open new therapeutic possibilities for clinical exploration.

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Authorship

Contribution: A.J.A. performed the experiments, analyzed and interpreted the data, performed data mining, prepared the figures, and cowrote the manuscript; E.C. performed the immunoblotting experiments; G. Sartori performed the silencing experiments; F.G. performed the assay for transposase-accessible chromatin for sequencing) data analyses and interpreted the data; L.C., L.T.d.B., and X.Y. performed data mining; S.N., G.F., F.F., C.T., and F. Spriano, performed the experiments; A.Z., F.M.R., R.B.P., and V.G. performed the flow cytometry analyses; A. Raimondi performed the electron microscopy; A.B. performed the variant calling; A. Rinaldi performed the genomics experiments; M.C.d.M. and M.E. performed the methylation profiling experiments and data mining; S.J. performed the enzyme-linked immunosorbent assay Luminex analyses; A.S. provided advice; Y.B.-D. and F. Simonetta provided reagents and advice; G. Stussi provided advice; J.R.B. and Q.P.-H. collected and characterized the tumor samples; E.Z. and D.R. codesigned the research and edited the manuscript; F.B. designed the research, interpreted the data, and cowrote the manuscript; and all authors approved the final manuscript.

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Footnotes

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The online version of this article contains a data supplement.

There is a [Blood Commentary](#) on this article in this issue.

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